

1. Fix the Syntax Errors:

```
x = 10
```

```
x+5
```

```
def g()
```

```
    global x
```

```
g()
```

```
print(x)
```

Question: Identify and fix the syntax errors in the above code.

2. What Will This Code Output?

```
x = 3
```

```
def update_x():
```

```
    global x
```

```
    x = x * 2
```

```
update_x()
```

```
print(x)
```

Question: Predict the output of the code.

3. Debugging a Function:

```
y = 4
```

```
def h():  
    global y  
    y = y + 10
```

```
h()  
print(y)
```

Question: Explain why the function works or doesn't work and provide the corrected code if needed.

4. Function Behavior:

```
x = 8
```

```
def increment():  
    global x  
    x += 1
```

```
increment()  
increment()  
print(x)
```

Question: What is the final value of x after the two calls to increment()?

5. Logical Error:

```
x = 7
```

```
def modify_x():  
    global x
```

```
x = x - 2
```

```
print(x)
```

Question: Will the above code produce any errors? If so, explain and correct them.

6. Define and Use Global Variable:

Write a Python program where:

- A global variable `z` is initially set to 100.
- A function `reduce_z` reduces the value of `z` by 25 each time it is called.
- Print the value of `z` before and after calling the function.